



Anomaly Detection in Spacecraft Telemetry: An MSL & SMAP Case Study

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Introduction

In the era of modern space exploration, the ability to monitor and analyze telemetry data in real-time is vital for mission success. Spacecraft such as MSL (Mars Science Laboratory) and SMAP (Soil Moisture Active Passive) generate massive volumes of telemetry data that reflect their operational health and environmental interactions.

This project focuses on the systematic analysis of these telemetry streams, with a primary objective to detect, categorize, and visualize anomalous behaviors. By leveraging Power BI, I have developed an interactive dashboard that transforms raw, complex data sets into actionable insights. This report outlines the methodology used to clean and structure the data, explains the choice of visual models, and presents the critical findings that highlight performance patterns, ultimately assisting in the proactive identification of operational risks and technical failures.



Methodology & Data Preparation

To ensure the accuracy and reliability of the analysis, the project followed a structured data pipeline approach:

- **Data Cleaning and Transformation:** The raw telemetry datasets were processed using **Power Query**. This involved handling missing values, standardizing data formats, and ensuring the consistency of anomaly timestamps and sequences. I performed data profiling to identify key metrics and eliminate redundant entries, preparing a clean foundation for the visualization layer.
- **Data Modeling:** I established robust relationships between the different data entities, specifically linking the spacecraft identifiers with their respective anomaly_sequences and class types. This relational modeling was essential to enable dynamic filtering and interactive drill-downs within the dashboard.
- **Visualization Strategy:** The design phase focused on three core visual components, each serving a specific analytical purpose:
 - **Distribution Analysis:** A Pie Chart was utilized to provide an immediate overview of the anomaly distribution between the MSL and SMAP spacecraft.
 - **Categorical Classification:** A Bar Chart was implemented to segment anomalies by their specific type (Point vs. Contextual), facilitating an understanding of failure patterns.

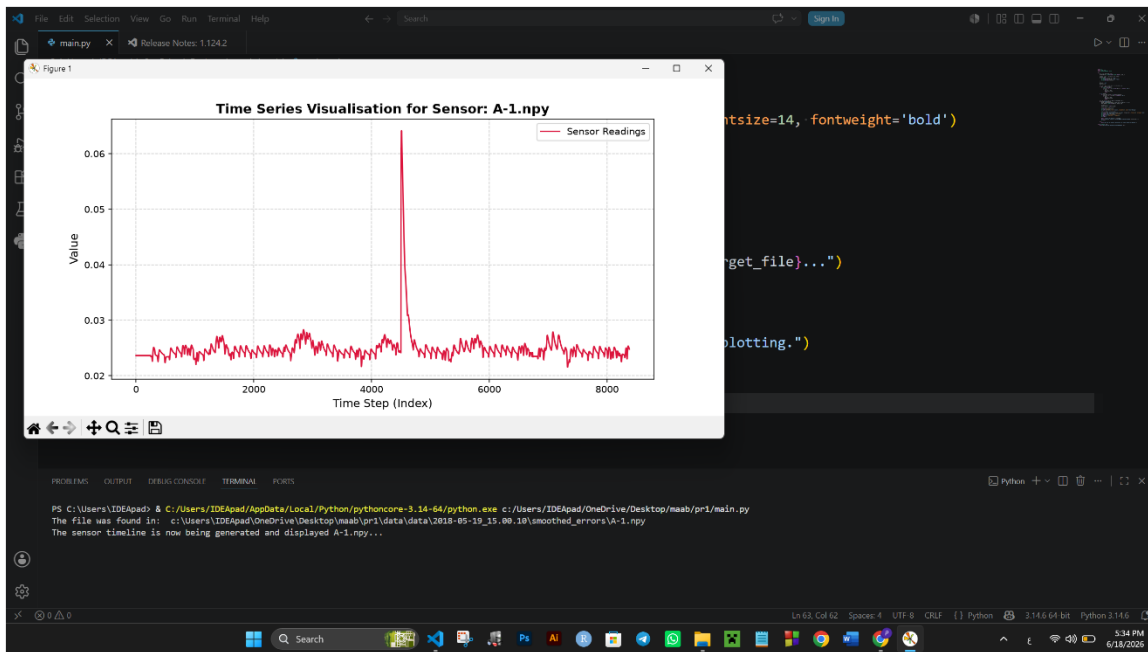


- **Trend & Pulse Monitoring:** A Scatter Plot (Anomaly Pulse Monitor) was developed to visualize anomaly intensity across event sequences. This allows for the tracking of "performance spikes" over time, making it easier to identify the most critical mission phases.
- **Interactivity & User Experience:** To empower the end-user, I integrated **Interactive Slicers**, allowing for seamless toggling between spacecraft. This provides the flexibility to isolate performance data for a specific mission while maintaining a holistic view of the overall telemetry health



Dashboard Architecture and Design Methodology

Exploratory Time-Series Analysis

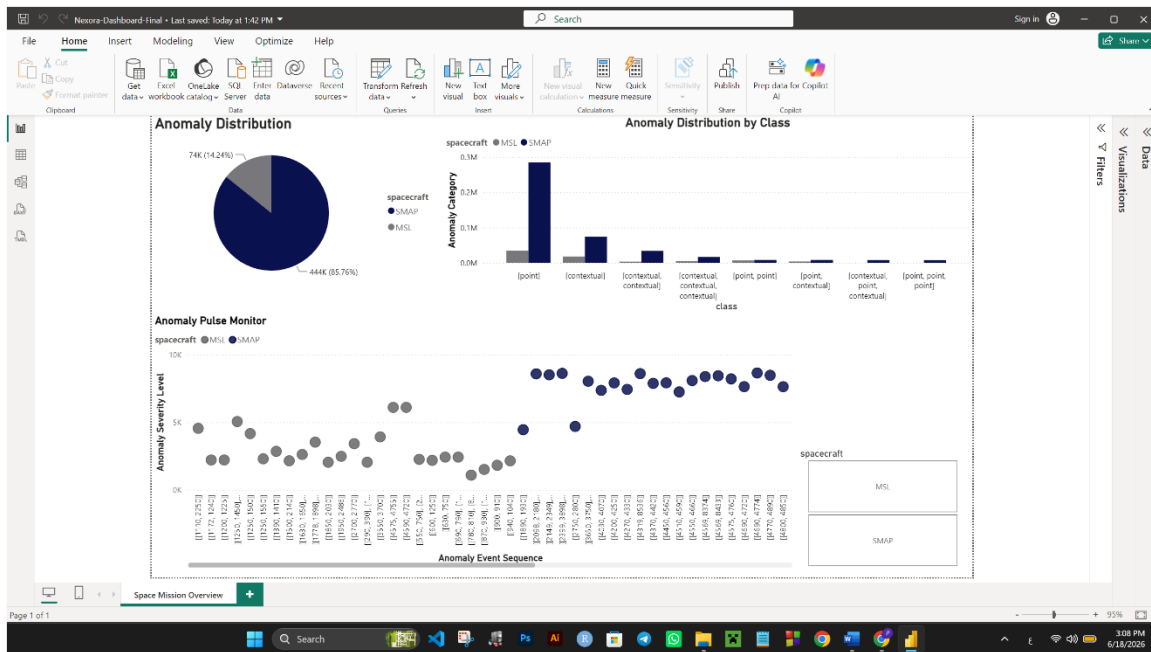


Before proceeding to interactive visualization, we conducted an initial time-series analysis for Sensor A-1 using Python. The objective of this phase was to understand the behavioral nature of the data and identify extreme outliers that indicate critical anomalies. As shown in the graph, there is a distinct 'peak spike' in the sensor readings at a specific time index, which signaled a significant anomaly requiring further investigation. This finding



served as the foundation for the categorical classification and dashboard integration in Power BI.

Dashboard Architecture



The dashboard architecture is built upon the **hierarchical data analysis methodology**. The visual journey begins with Key Performance Indicators (KPIs) and high-level summaries (Overview), progressively guiding the user toward deep-dive predictive and temporal analyses. This design ensures that the user can access critical information in under three clicks, while the integrated self-filtering feature enhances the precision of comparative analysis between space assets.



1. Level I: High-Level Overview

The Pie Chart serves as the primary **entry point**. It provides an immediate answer to the most critical question: *"Which spacecraft accounts for the majority of anomalies?"* Without this summary, the user would be overwhelmed by raw data.

2. Level II: Granular Categorization

The Bar Chart addresses the *"Why?"* by breaking down complex datasets into specific anomaly classes (e.g., Point vs. Contextual). This level of analysis provides the actionable insights necessary for engineers to implement corrective measures.

3. Level III: Temporal Analysis (Pattern Recognition)

The Scatter Plot acts as the **"heart" of the dashboard**. Here, the focus shifts from "what happened" to "when it happened and at what intensity." This visualization enables the identification of recurrent patterns, such as data point clusters within specific sequences, which are vital for failure diagnostics.

4. Level IV: Interactivity and Control

The Slicer, configured as interactive buttons, functions as the **"search engine"** of the dashboard. It grants users the authority to isolate and focus on specific data segments, facilitating a tailored and efficient analytical experience.



Dashboard Architecture and Design Methodology

1. **SMAP Dominance:** The data indicates that the SMAP spacecraft accounts for the majority of detected anomalies (85.76%). This suggests that either SMAP operates in a more complex environment or requires more frequent sensor maintenance compared to the MSL unit.
2. **Anomalous Patterning:** The categorical analysis reveals that the "Point" anomaly class is the most frequent across both spacecraft. This implies that most failures are transient, momentary events rather than continuous structural breakdowns, directing engineers to focus on sensor stability.
3. **Anomaly Clustering:** The "Anomaly Pulse Monitor" demonstrates that anomalies are not randomly distributed but rather concentrated in specific event sequences. These clusters indicate a strong correlation between anomaly frequency and specific operational mission phases



Technical Recommendations

Based on the insights derived from our telemetry analysis, we recommend the following operational steps to optimize spacecraft performance:

1. **Predictive Maintenance Protocol:** Given the prevalence of "Point" anomalies in the SMAP unit, we recommend updating real-time filtering algorithms to minimize false positives caused by instantaneous sensor fluctuations.
2. **Root Cause Analysis for Critical Sequences:** Conduct a deep-dive investigation into the log files for event sequences showing high anomaly intensity (peaks) to determine if these failures are triggered by environmental conditions or mechanical component stress.
3. **Enhanced Monitoring Strategy:** Implement a "Smart Alert" system based on specific severity thresholds observed in the Pulse Monitor scatter plot. This will ensure timely intervention before "Contextual" anomalies escalate into critical failures.



Challenges & Learnings

Throughout the development of this project, we encountered technical challenges that sharpened our data engineering and analytical skills:

1. **Data Cleaning & Preprocessing:** The telemetry datasets contained significant "noise" and inconsistencies. I learned how to clean and structure these complex logs to ensure the reliability of the final visual analysis.
2. **Visual Design Trade-offs:** The core challenge was displaying massive amounts of data without overwhelming the user. I mastered the principle of "Visual Hierarchy," utilizing interactive Slicers to streamline the user experience (UX) without sacrificing depth.
3. **Data Storytelling:** I learned that an analyst's role extends beyond building charts; it is about uncovering the story hidden within the data and communicating actionable insights to stakeholders effectively.



Conclusion

This project demonstrates the power of data in anticipating the performance of mission-critical systems. Through the 'Nexora' dashboard, we have moved beyond simple monitoring to create a strategic tool that supports decision-making and contributes to extending the operational lifespan of space assets. The success of this system in translating complex telemetry into clear insights reaffirms that the future of space exploration relies fundamentally on data intelligence. This project is just the beginning, as we aim to evolve it into an AI-driven predictive system capable of detecting anomalies before they occur